

Section A – Basic Functions and Numbers

1. Even or Odd

Write a function `check_even_odd(n)` that accepts a number and prints whether the number is even or odd.

Concepts: function, if-else, %

2. Positive, Negative or Zero

Write a function `check_number(n)` that accepts a number and prints whether it is positive, negative, or zero.

Concepts: function, if-elif-else

3. Find Largest of Two Numbers

Write a function `find_largest(a, b)` that accepts two numbers and returns the largest number.

Concepts: function, return, conditional statement

4. Find Largest of Three Numbers

Write a function `find_largest(a, b, c)` that finds the largest among three numbers.

Restriction: Do not use `max()`.

5. Sum of Natural Numbers

Write a function `sum_natural(n)` that calculates the sum of numbers from 1 to n.

Restriction: Use a for loop.

Example:

Enter n: 10

Sum = 55

6. Multiplication Table

Write a function `multiplication_table(n)` that prints the multiplication table of a given number from 1 to 10.

Concepts: for, range()

7. Factorial Using Function

Write a function `factorial(n)` that calculates the factorial of a number.

Restriction: Use a loop. Do not use recursion.

8. Count Digits

Write a function `count_digits(n)` that counts the number of digits in a number.

Restriction: Use a while loop.

9. Reverse a Number

Write a function `reverse_number(n)` that reverses a given number.

Example:

Input : 12345

Output : 54321

Concepts: while, %, //

10. Prime Number

Write a function `check_prime(n)` that checks whether a number is prime.

Restriction: Use a loop and do not use any external library.

Section B – String-Based Assignments

11. Count Characters in a String

Write a function `count_characters(text)` that counts the number of characters in a string.

Restriction: Do not use `len()`.

Concepts: for loop, counter

12. Count Vowels

Write a function `count_vowels(text)` that counts the number of vowels in a string.

Example:

Input : education

Output : 5

Restriction: Use a loop.

13. Count Consonants

Write a function `count_consonants(text)` that counts the number of consonants in a string.

Assume: Input contains alphabetic characters and spaces.

14. Count Vowels and Consonants

Write a function `count_vowels_consonants(text)` that separately counts vowels and consonants.

Example:

Input : Python Programming

Vowels : 4

Consonants : 14

15. Reverse a String

Write a function `reverse_string(text)` that reverses a string.

Restriction: Demonstrate both approaches:

Approach 1: Using a loop

Approach 2: Using string slicing

`text[::-1]`

16. Check Palindrome String

Write a function `check_palindrome(text)` that checks whether a string is a palindrome.

Example:

madam → Palindrome

hello → Not Palindrome

17. Count Words

Write a function `count_words(text)` that counts the number of words in a sentence.

Restriction: Do not use `split()`.

Example:

Input:

Python is easy to learn

Output:

5

Hint: Count spaces carefully.

18. Find Frequency of a Character

Write a function `character_frequency(text, ch)` that counts how many times a particular character occurs.

Restriction: Do not use `count()`.

Example:

Text : programming

Character : g

Frequency : 2

19. Remove Spaces

Write a function `remove_spaces(text)` that removes all spaces from a string.

Restriction: Use a loop.

Example:

Input : Python Programming Language

Output : PythonProgrammingLanguage

20. Convert Lowercase to Uppercase

Write a function `convert_uppercase(text)` that converts a string to uppercase.

Task: First solve using `upper()` and then demonstrate a loop-based solution using character processing.

Section C – String + Loop Problems

21. Count Uppercase and Lowercase Characters

Write a function `count_case(text)` that counts:

- uppercase characters
- lowercase characters
- digits
- spaces

Example:

Input: Python123 ABC

Uppercase : 3

Lowercase : 6

Digits : 3

Spaces : 1

22. Find First Character

Write a function `first_character(text)` that prints the first character of a string.

Restriction: Do not directly use `text[0]`.

Use a loop and break.

23. Find Last Character

Write a function `last_character(text)` that prints the last character of a string.

Restriction: Demonstrate:
text[-1]
and a loop-based approach.

24. Print Each Character

Write a function display_characters(text) that prints every character of a string on a separate line.

Example:

Input:

PYTHON

Output:

P

Y

T

H

O

N

Use: for loop.

25. Print Characters with Position

Write a function display_position(text) that displays every character along with its position.

Example:

Input:

JAVA

Output:

Position 0 : J

Position 1 : A

Position 2 : V

Position 3 : A

Concepts: range(), indexing

26. Remove Vowels

Write a function remove_vowels(text) that removes all vowels from a string.

Example:

Input:

Python Programming

Output:

Pythn Prgrmmng

Restriction: Use a loop. Do not use replace().

27. Find Longest Word

Write a function `find_longest_word(text)` that finds the longest word in a sentence.

Restriction:

- Do not use dictionaries.
 - Do not use `max()`.
 - Do not use advanced collections.
 - Use loops and string processing.
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28. Count Occurrence of Each Vowel

Write a function that separately counts:

a

e

i

o

u

Example:

Input:

education

Output:

a = 1

e = 1

i = 1

o = 1

u = 1

Restriction: Use loops and conditions.

Section D – Loops, Patterns and Menus

29. Star Pattern

Write a function `star_pattern(n)` that prints:

*

**

For n = 5.

Concepts: nested loops.

30. Number Pattern

Write a function `number_pattern(n)` that prints:

1

12

123

1234

12345

Concepts: nested for loops.
